# Digital payment adoption across Indian states, 2018–2026: building the state panel from PhonePe Pulse

BTP study, Part 1

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**Abstract**

Table 1: PhonePe Pulse files used in the study. Unit: state-quarter. Sample: 36 raw state and union-territory names, 34 quarters, 2018Q1–2026Q2. Source: PhonePe [2], pinned commit.

| Folder (under `data/`) | Content and use | Items |
|---|---|---|
| `map/transaction/hover` | Transaction count and amount by state | 34 quarter files; 3,570 rows in the interim file |
| `map/user/hover` | Registered users by state | 34 quarter files; in the same interim file |
| `aggregated/transaction/country/india/state` | Transaction count by type: P2P, Retail, Utility | 36 names × 34 quarters; 3,570 rows in the interim file |
| `aggregated/transaction/country` and `aggregated/user/country` | National totals, used for checks only | 34 quarter files each |

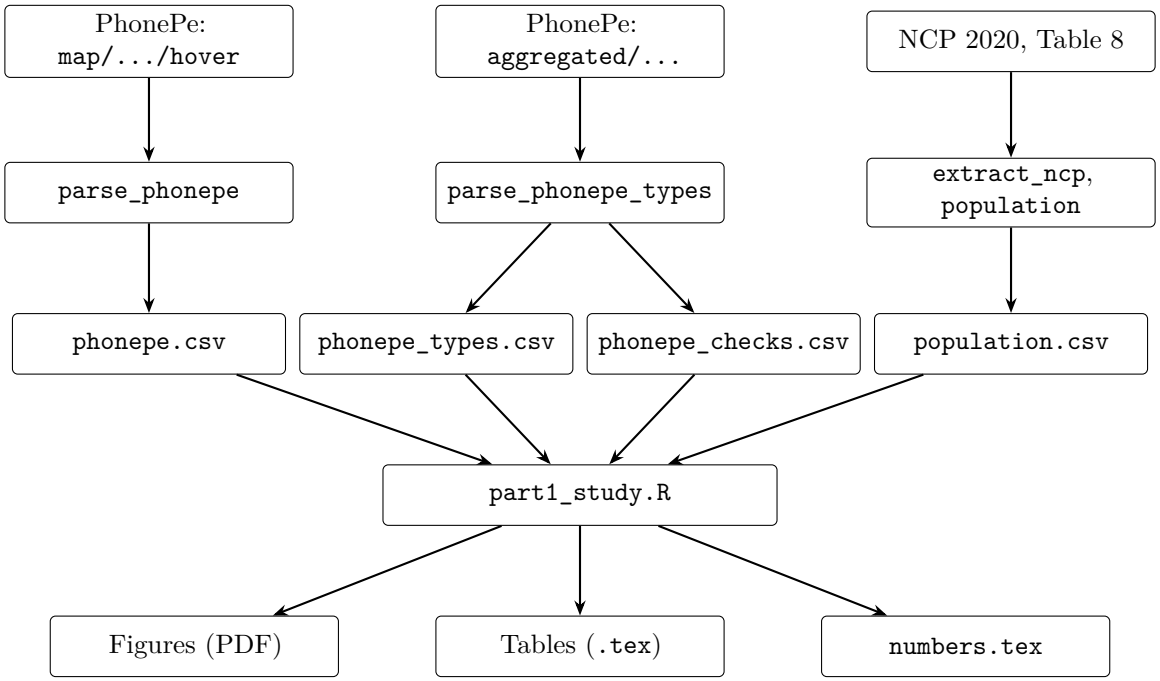



Figure 1: Data flow from the three raw sources to the study outputs. Raw files: PhonePe [2] and National Commission on Population [1]. Each Python module writes one or two interim files with the columns `state_id`, `period`, `variable`, `value` and `source`, except the check file.

Table 2: Data checks. Unit: the item named in each row (state-quarters, state-quarter-type entries, quarters, state rows or panel cells). Sample: all 35 state units, 34 quarters. The last column is the largest absolute difference for the PhonePe rows and the ratio of the state sum to the India row for the NCP rows. Source: PhonePe [2], National Commission on Population [1].

| Check | Items | Flagged | Difference or ratio |
|---|---|---|---|
| PhonePe: sum of the three type counts against the state count (state-quarters) | 1,224 | 0 | 0 |
| PhonePe: absent type entries set to zero (state-quarter-type entries). Flagged: Lakshadweep, 2018Q1, Utility | 3,672 | 1 | 0 |
| PhonePe: national count file against the sum of the state counts (quarters) | 34 | 0 | 0 |
| PhonePe: national user file against the sum of the state user counts (quarters) | 34 | 0 | 0 |
| NCP 2020, 2018: sum of the state rows divided by the India row (state rows) | 37 | – | 1.0000015 |
| NCP 2020, 2027: sum of the state rows divided by the India row (state rows) | 37 | – | 0.9999993 |
| Population: missing cells in the panel (35 ids × 34 quarters) | 1,190 | 0 | – |

Table 3: National annual totals and growth. Unit: count in billion, amount in Rs. lakh crore (nominal), users at year end in million, count per person per year, average value in Rs., growth of count in percent. Sample: all 35 state units. Period: calendar years 2018–2026 (2026 has two quarters). Source: PhonePe [2]; population from National Commission on Population [1].

| Year | Count (billion) | Amount (Rs. lakh crore) | Users at year end (million) | Count per person | Average value (Rs.) | Growth of count (percent) |
|---|---|---|---|---|---|---|
| 2018 | 1.11 | 1.63 | 103.0 | 0.8 | 1,467 | – |
| 2019 | 4.12 | 6.29 | 179.9 | 3.1 | 1,525 | 271.9 |
| 2020 | 8.02 | 14.65 | 266.2 | 5.9 | 1,827 | 94.5 |
| 2021 | 18.45 | 33.09 | 351.7 | 13.5 | 1,793 | 130.0 |
| 2022 | 38.64 | 62.97 | 436.0 | 28.0 | 1,630 | 109.4 |
| 2023 | 64.30 | 94.60 | 513.0 | 46.2 | 1,471 | 66.4 |
| 2024 | 99.33 | 129.71 | 598.7 | 70.8 | 1,306 | 54.5 |
| 2025 | 128.54 | 155.38 | 676.8 | 90.8 | 1,209 | 29.4 |
| 2026 (Q1–Q2) | 74.92 | 88.94 | 711.7 | 52.6 | 1,187 | – |

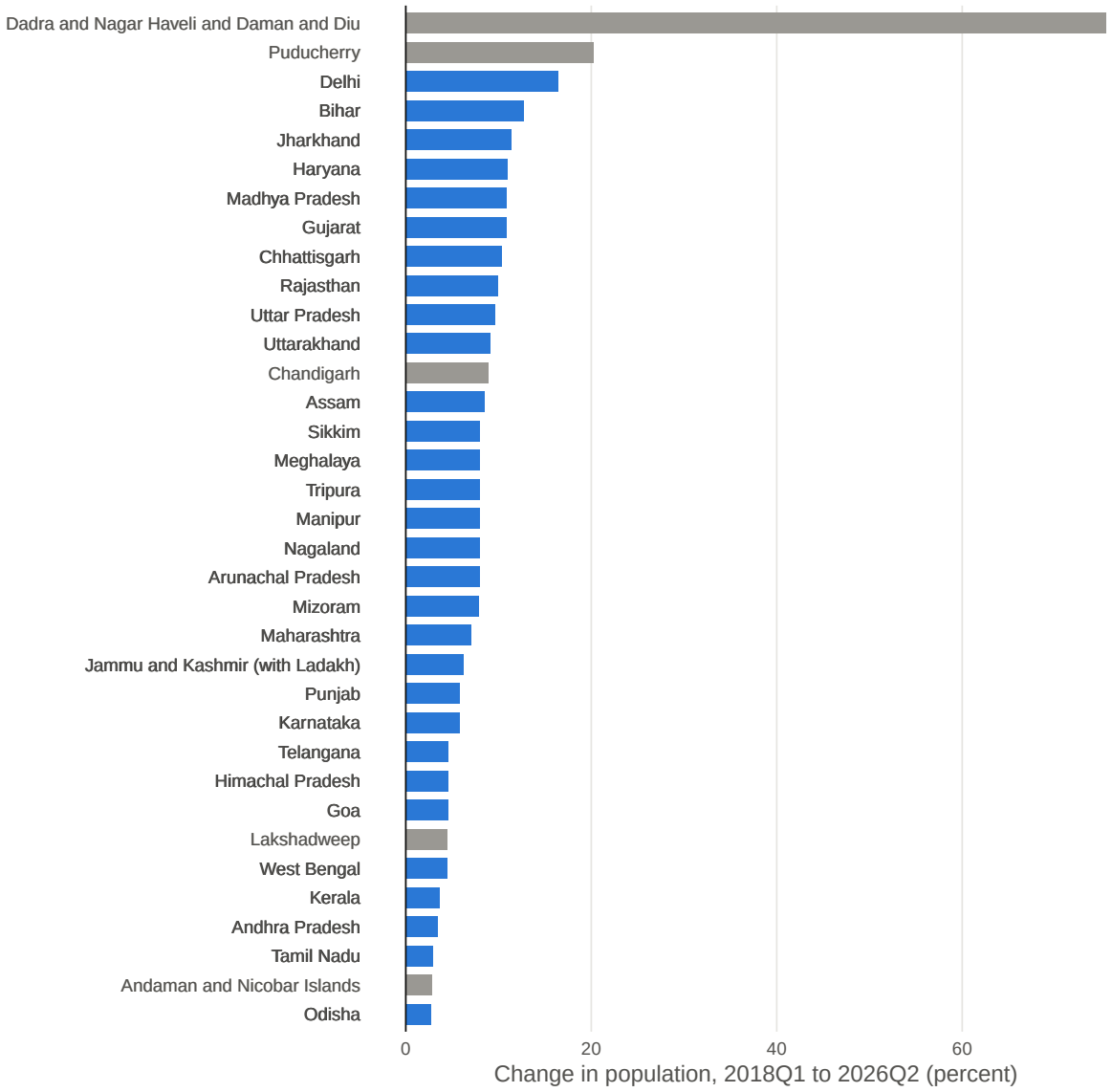



Figure 2: Population growth by state unit, 2018Q1 to 2026Q2. Unit: percent change. Sample: all 35 state units; the other union territories are marked. Source: interpolation of the NCP 2020 projections [1].

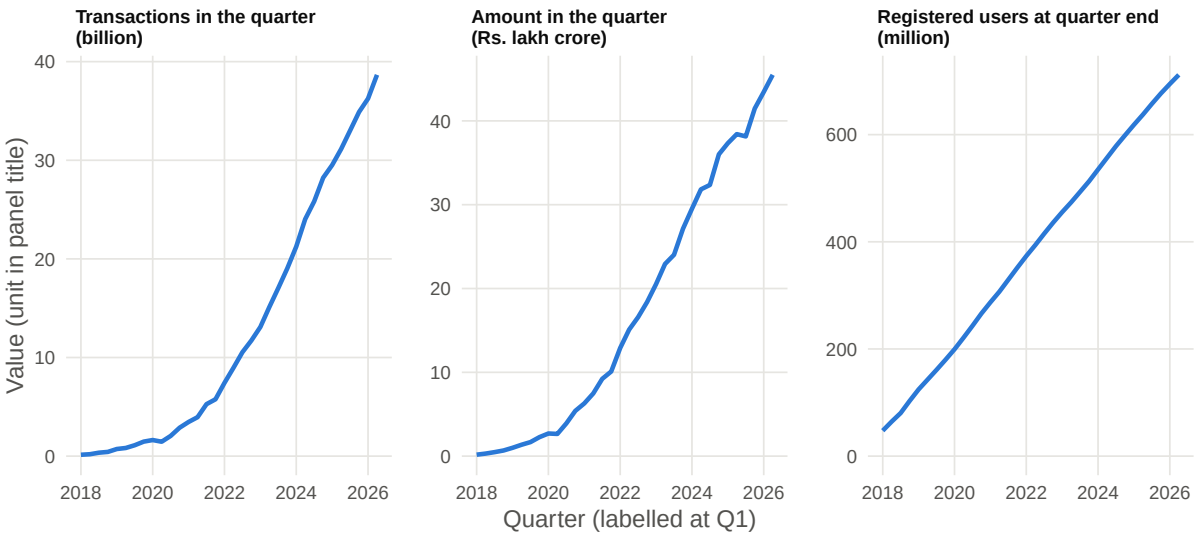



Figure 3: National transaction count (billion), transaction amount (Rs. lakh crore) and registered users (million), by quarter. Unit: national total of all 35 state units. Period: 2018Q1–2026Q2. Source: PhonePe [2].

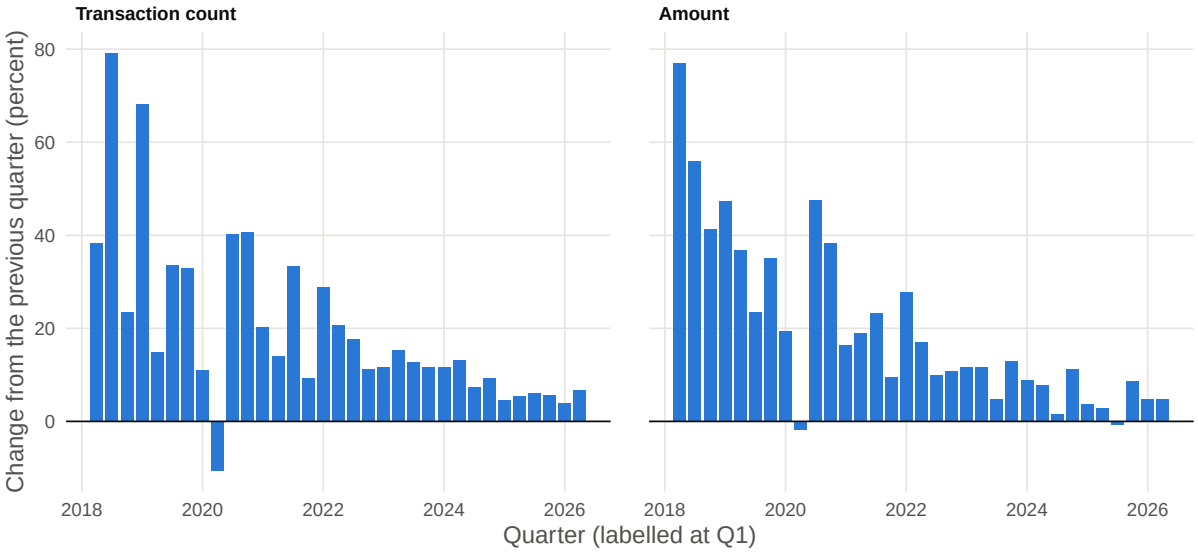



Figure 4: Quarter-on-quarter growth of the national transaction count and amount. Unit: percent change from the previous quarter. Sample: national total of all 35 state units. Period: 2018Q2–2026Q2. Source: PhonePe [2].

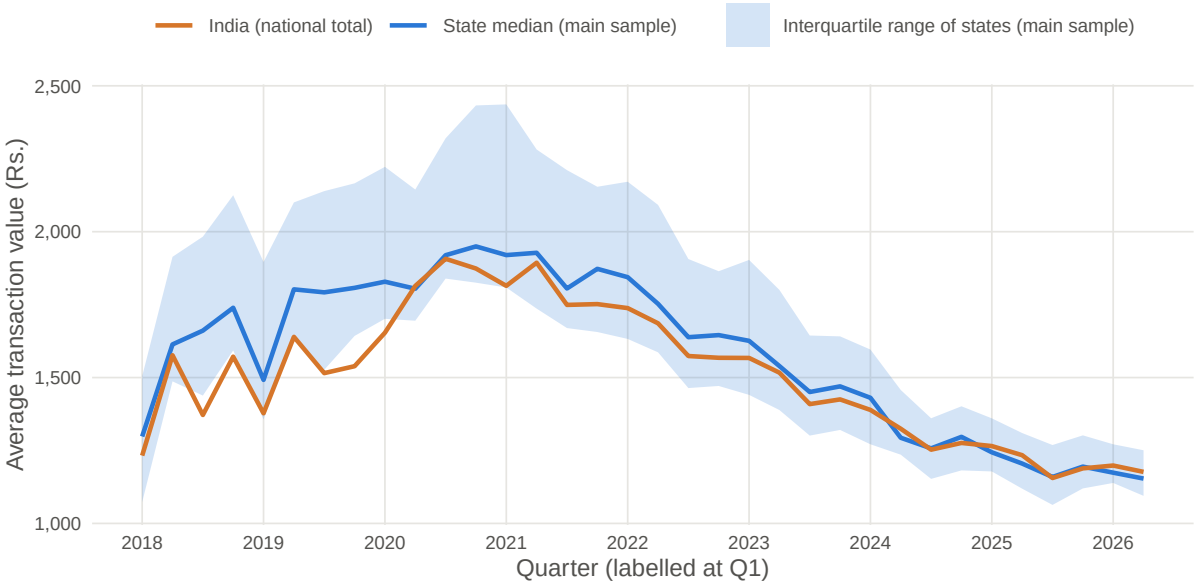



Figure 5: Average transaction value. Unit: Rs. per transaction (nominal), amount divided by count. Sample: national value, and median and interquartile range across the 30 main-sample states. Period: 2018Q1–2026Q2. Source: PhonePe [2].

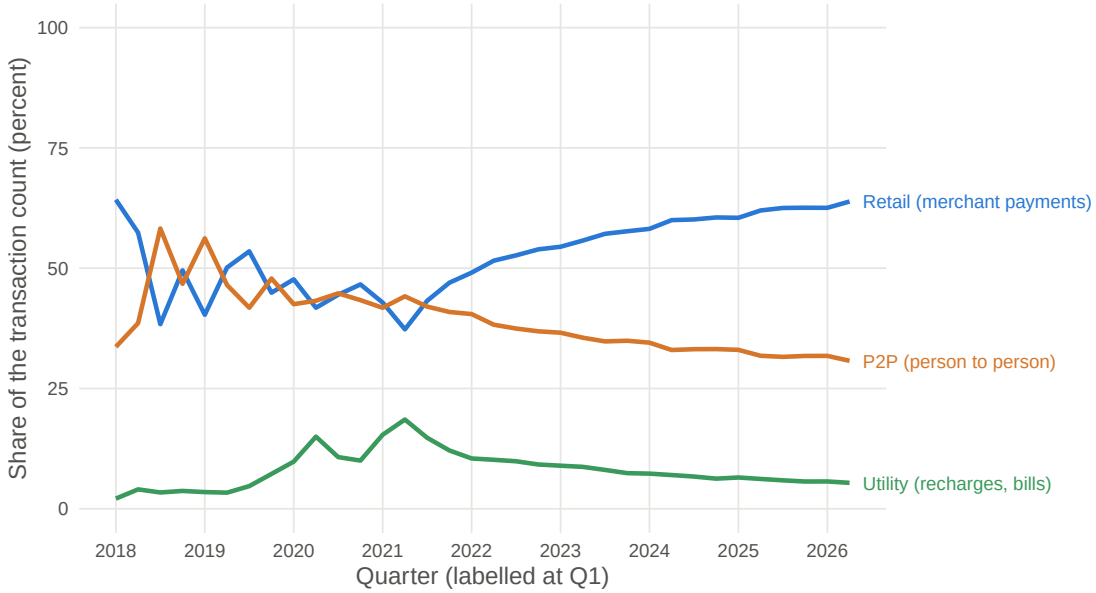



Figure 6: National shares of the transaction count by type: P2P, Retail and Utility. Unit: percent of the national count. Sample: national total of all 35 state units. Period: 2018Q1–2026Q2. Source: PhonePe [2].

# 1 Introduction

## 1.1 The question and the data gap

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## 1.3 Aim and contents of this part

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## 4.3 Average transaction value

## 4.4 Transaction types

# 5 Adoption across states

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## 5.3 Rank persistence

## 5.4 Reach and intensity

## 5.5 Decomposition of the variance

## 5.6 Retail share

# 6 Limitations of the data

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# References



[1] National Commission on Population. Report of the technical group on population projections:

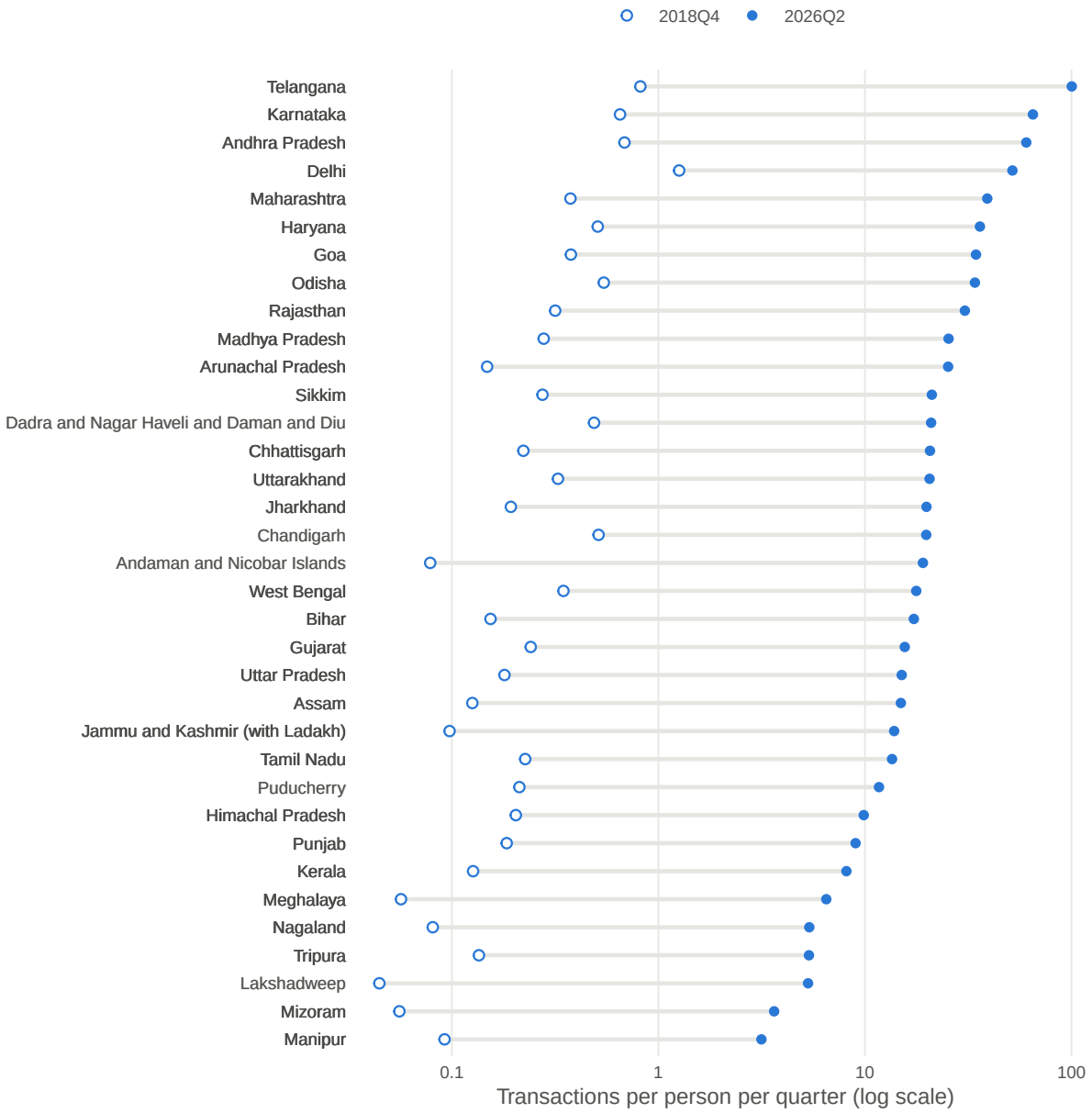



Figure 7: Transaction count per person per quarter, by state unit, in 2018Q4 and 2026Q2. Unit: transactions per person per quarter. Sample: all 35 state units; the other union territories are marked. Source: PhonePe [2]; population from National Commission on Population [1].

Table 4: State values in 2026Q2. Unit: population in million; count per person per quarter; amount per person in Rs. (nominal); users per 1000 persons; transactions per user; average value in Rs.; retail share in percent. Sample: the 30 main-sample states, sorted by count per person. Source: PhonePe [2]; population from National Commission on Population [1].

| Name | Population (million) | Count per person | Amount per person (Rs.) | Users per 1000 persons | Count per user | Average value (Rs.) | Retail share (percent) |
|---|---|---|---|---|---|---|---|
| Telangana | 38.66 | 100.3 | 132,629 | 943 | 106.3 | 1,322 | 66.0 |
| Karnataka | 69.05 | 65.1 | 76,649 | 761 | 85.5 | 1,178 | 67.5 |
| Andhra Pradesh | 53.73 | 60.4 | 86,051 | 639 | 94.6 | 1,424 | 61.0 |
| Delhi | 22.64 | 51.8 | 51,967 | 939 | 55.1 | 1,004 | 68.5 |
| Maharashtra | 129.52 | 39.1 | 42,197 | 649 | 60.2 | 1,079 | 67.4 |
| Haryana | 31.38 | 36.0 | 40,263 | 728 | 49.5 | 1,117 | 67.9 |
| Goa | 1.60 | 34.5 | 43,161 | 842 | 41.0 | 1,251 | 67.9 |
| Odisha | 44.69 | 34.1 | 37,283 | 505 | 67.4 | 1,094 | 61.2 |
| Rajasthan | 83.82 | 30.5 | 38,052 | 521 | 58.4 | 1,249 | 64.2 |
| Madhya Pradesh | 89.89 | 25.4 | 28,572 | 446 | 57.0 | 1,125 | 63.8 |
| Arunachal Pradesh | 1.61 | 25.3 | 29,550 | 592 | 42.7 | 1,169 | 60.8 |
| Sikkim | 0.71 | 21.1 | 25,729 | 574 | 36.7 | 1,221 | 58.3 |
| Dadra and Nagar Haveli and Daman and Diu† | 1.58 | 20.9 | 16,808 | 434 | 48.2 | 804 | 69.3 |
| Chhattisgarh | 31.29 | 20.7 | 21,684 | 422 | 49.0 | 1,050 | 62.3 |
| Uttarakhand | 12.02 | 20.6 | 21,235 | 562 | 36.6 | 1,033 | 66.4 |
| Jharkhand | 41.07 | 19.9 | 22,132 | 384 | 51.6 | 1,115 | 61.2 |
| Chandigarh† | 1.27 | 19.8 | 19,877 | 653 | 30.3 | 1,004 | 71.0 |
| Andaman and Nicobar Islands† | 0.41 | 19.1 | 25,704 | 543 | 35.1 | 1,347 | 63.4 |
| West Bengal | 100.60 | 17.7 | 21,986 | 430 | 41.2 | 1,241 | 56.4 |
| Bihar | 132.70 | 17.2 | 20,181 | 338 | 51.1 | 1,170 | 56.6 |
| Gujarat | 74.28 | 15.6 | 17,471 | 486 | 32.1 | 1,121 | 65.4 |
| Uttar Pradesh | 243.31 | 15.1 | 16,188 | 372 | 40.5 | 1,075 | 62.6 |
| Assam | 36.79 | 14.9 | 14,126 | 383 | 39.0 | 946 | 57.3 |
| Jammu and Kashmir (with Ladakh) | 14.22 | 13.8 | 15,360 | 415 | 33.4 | 1,109 | 62.3 |
| Tamil Nadu | 77.57 | 13.5 | 16,974 | 519 | 26.1 | 1,255 | 63.6 |
| Puducherry† | 1.77 | 11.7 | 13,528 | 409 | 28.6 | 1,157 | 69.1 |
| Himachal Pradesh | 7.59 | 9.9 | 10,819 | 474 | 20.8 | 1,096 | 64.1 |
| Punjab | 31.36 | 9.0 | 11,788 | 450 | 20.0 | 1,309 | 62.7 |
| Kerala | 36.23 | 8.1 | 9,260 | 450 | 18.1 | 1,139 | 65.1 |
| Meghalaya | 3.45 | 6.5 | 7,059 | 238 | 27.3 | 1,086 | 48.4 |
| Nagaland | 2.30 | 5.4 | 7,123 | 262 | 20.5 | 1,325 | 49.2 |
| Tripura | 4.27 | 5.4 | 6,265 | 291 | 18.4 | 1,170 | 56.7 |
| Lakshadweep† | 0.07 | 5.3 | 5,294 | 263 | 20.2 | 998 | 67.0 |
| Mizoram | 1.27 | 3.6 | 4,769 | 274 | 13.3 | 1,313 | 48.2 |
| Manipur | 3.32 | 3.2 | 5,118 | 257 | 12.2 | 1,623 | 43.5 |

† Union territory outside the main sample.

Table 5: Concentration of activity in the five states with the largest transaction count. Unit: share of the national total, in percent. Sample: all 35 state units. Periods: 2018Q4 and 2026Q2. Source: PhonePe [2]; population from National Commission on Population [1].

| Share of the top five (by count) | 2018Q4 | 2026Q2 |
|---|---|---|
| Transaction count (percent) | 46.2 | 52.6 |
| Amount (percent) | 44.1 | 53.7 |
| Registered users (percent) | 43.7 | 41.9 |
| Population (percent) | 42.2 | 37.5 |
| States needed for half of all transactions | 6 | 5 |

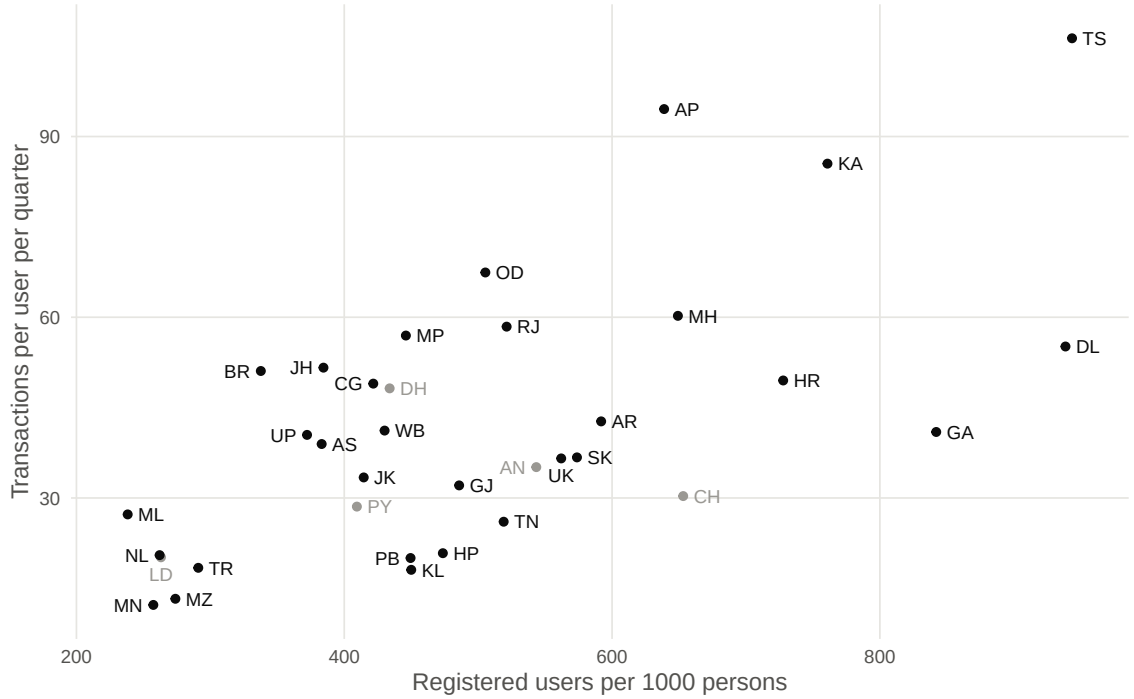



Figure 8: Reach against intensity in 2026Q2. Unit: registered users per 1000 persons (reach) and transactions per registered user per quarter (intensity). Sample: all 35 state units; the other union territories are marked. Source: PhonePe [2]; population from National Commission on Population [1].

Table 6: Variance decomposition of the log of transactions per person: $y = \ln(C/P)$, reach $r = \ln(U/P)$ and intensity $i = \ln(C/U)$. Unit: variance of logs across states; contributions as shares of $\mathrm{var}(y)$. Sample: the 30 main-sample states. Periods: 2018Q4 and 2026Q2. Source: PhonePe [2]; population from National Commission on Population [1].

| | 2018Q4 | 2026Q2 |
|---|---|---|
| $n$ | 30 | 30 |
| $\mathrm{var}(y)$ | 0.595 | 0.723 |
| $\mathrm{var}(r)$ | 0.397 | 0.140 |
| $\mathrm{var}(i)$ | 0.069 | 0.305 |
| $2\,\mathrm{cov}(r, i)$ | 0.129 | 0.278 |
| Reach contribution | 0.776 | 0.386 |
| Intensity contribution | 0.224 | 0.614 |

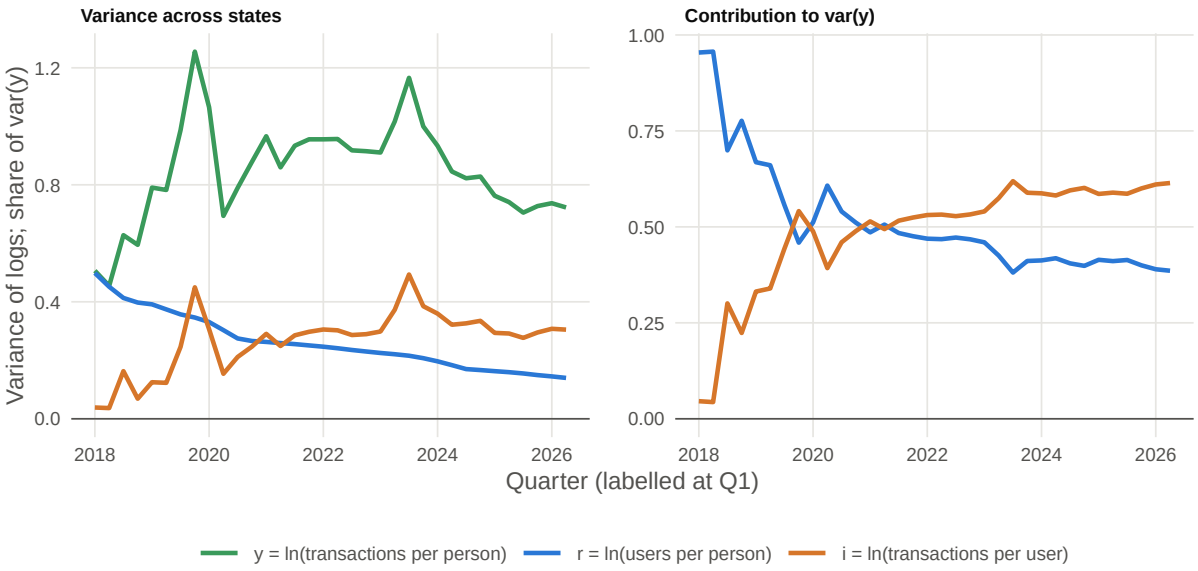



Figure 9: Contributions of reach and intensity to the cross-state variance of the log of transactions per person. Unit: share of the variance. Sample: the 30 main-sample states. Period: 2018Q1–2026Q2. Source: PhonePe [2]; population from National Commission on Population [1].

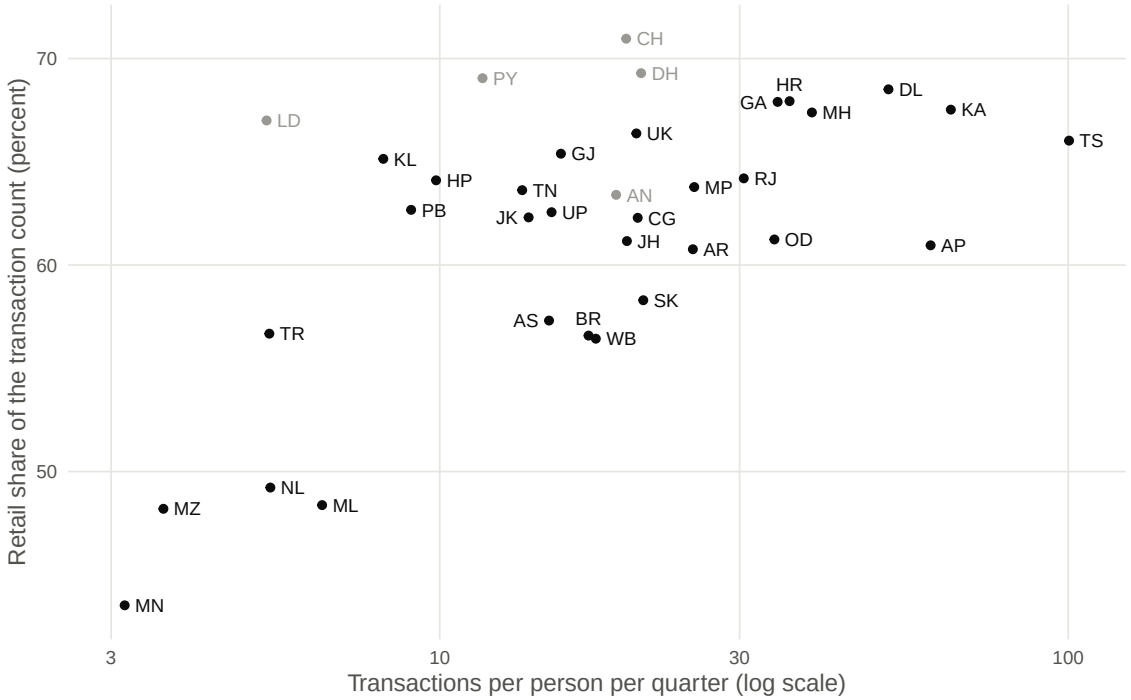



Figure 10: Retail share of the transaction count against transactions per person in 2026Q2. Unit: percent of the state count; transactions per person per quarter. Sample: the 30 main-sample states. Source: PhonePe [2]; population from National Commission on Population [1].

India and states 2011–2036. Technical report, Ministry of Health and Family Welfare, Government of India, July 2020. URL https://nhm.gov.in/New_Updates_2018/Report_Population_Projection_2019.pdf. Table 8, projected total population as on 1 March.

[2] PhonePe. PhonePe Pulse data repository. https://github.com/PhonePe/pulse, commit 257bf38e3095020f9f516c4c72c10bdc348781b1, 2026. CDLA-Permissive-2.0; data 2018Q1–2026Q2; accessed 1 November 2026.

# A The 35 state units

The names in the last two columns are the raw names after the normalisation of the state crosswalk: lower case, with "and" in place of "&". A semicolon separates the raw names that the crosswalk sums into one unit.

Table 7: The 35 state units

| Name | state_id | Code | Main | PhonePe raw names | NCP raw names |
|---|---|---|---|---|---|
| Andaman and Nicobar Islands | andaman_nicobar | AN | No | andaman and nicobar islands | andaman and nicobar islands |
| Andhra Pradesh | andhra_pradesh | AP | Yes | andhra pradesh | andhra pradesh |
| Arunachal Pradesh | arunachal_pradesh | AR | Yes | arunachal pradesh | arunachal pradesh |
| Assam | assam | AS | Yes | assam | assam |
| Bihar | bihar | BR | Yes | bihar | bihar |
| Chandigarh | chandigarh | CH | No | chandigarh | chandigarh |
| Chhattisgarh | chhattisgarh | CG | Yes | chhattisgarh | chhattisgarh |
| Dadra and Nagar Haveli and Daman and Diu | dnh_dd | DH | No | dadra and nagar haveli and daman and diu | dadra and nagar haveli; daman and diu |
| Delhi | delhi | DL | Yes | delhi | nct of delhi |
| Goa | goa | GA | Yes | goa | goa |
| Gujarat | gujarat | GJ | Yes | gujarat | gujarat |
| Haryana | haryana | HR | Yes | haryana | haryana |
| Himachal Pradesh | himachal_pradesh | HP | Yes | himachal pradesh | himachal pradesh |
| Jammu and Kashmir (with Ladakh) | jammu_kashmir | JK | Yes | jammu and kashmir; ladakh | jammu and kashmir; ladakh |
| Jharkhand | jharkhand | JH | Yes | jharkhand | jharkhand |
| Karnataka | karnataka | KA | Yes | karnataka | karnataka |
| Kerala | kerala | KL | Yes | kerala | kerala |
| Lakshadweep | lakshadweep | LD | No | lakshadweep | lakshadweep |
| Madhya Pradesh | madhya_pradesh | MP | Yes | madhya pradesh | madhya pradesh |
| Maharashtra | maharashtra | MH | Yes | maharashtra | maharashtra |
| Manipur | manipur | MN | Yes | manipur | manipur |
| Meghalaya | meghalaya | ML | Yes | meghalaya | meghalaya |
| Mizoram | mizoram | MZ | Yes | mizoram | mizoram |
| Nagaland | nagaland | NL | Yes | nagaland | nagaland |
| Odisha | odisha | OD | Yes | odisha | odisha |
| Puducherry | puducherry | PY | No | puducherry | puducherry |
| Punjab | punjab | PB | Yes | punjab | punjab |
| Rajasthan | rajasthan | RJ | Yes | rajasthan | rajasthan |
| Sikkim | sikkim | SK | Yes | sikkim | sikkim |
| Tamil Nadu | tamil_nadu | TN | Yes | tamil nadu | tamil nadu |
| Telangana | telangana | TS | Yes | telangana | telangana |

| Name | state_id | Code | Main | PhonePe raw names | NCP raw names |
|---|---|---|---|---|---|
| Tripura | tripura | TR | Yes | tripura | tripura |
| Uttar Pradesh | uttar_pradesh | UP | Yes | uttar pradesh | uttar pradesh |
| Uttarakhand | uttarakhand | UK | Yes | uttarakhand | uttarakhand |
| West Bengal | west_bengal | WB | Yes | west bengal | west bengal |

# B Reproduction

The commands below run from the repository root. They fetch the raw files, build the interim files, write the figures, tables and numbers, and build this document.

```
make data analysis report
```

The PhonePe Pulse files are at the commit `257bf38e3095020f9f516c4c72c10bdc348781b1`. The study reads these input files:

- `data/interim/phonepe.csv`
- `data/interim/phonepe_types.csv`
- `data/interim/phonepe_checks.csv`
- `data/interim/population.csv`
- `src/R/lib/state_labels.csv`

The script `src/R/part1_study.R` writes the figures to `output/study/part1/figures/`, the tables to `output/study/part1/tables/` and the numbers to `output/study/part1/numbers.tex`. The file `numbers.tex` is generated. Do not edit it. A number in the text comes from this file with the command `\val`, and an unknown key stops the compile.